TALENTMATCH AI

Intelligent Candidate Discovery & Hybrid Ranking Engine

Hackathon Track: Data & AI Challenge | **Approach Document & Presentation Deck**

The Problem: Limitations of Legacy Filtering

Traditional Applicant Tracking Systems (ATS) and filtering tools fail both organizations and candidates because they operate on shallow textual patterns.

❌ Blind Keyword Matching

Legacy platforms filter resumes purely on exact character strings. If a candidate uses an equivalent synonym or describes an architectural pattern without naming the exact tool keyword, they are instantly dropped and missed entirely.

❌ Vulnerability to Keyword Stuffing

Unqualified profiles easily exploit traditional string filters by pasting massive blocks of buzzwords into their metadata. Legacy rankers mistakenly score these "stuffed" profiles higher than genuine talent, wasting thousands of recruiter hours.

The Hackathon Reality: Processing 100,000 deep profiles with complex parameters cannot be done using raw LLM loops or basic string counting without triggering either massive computation timeouts or incorrect candidate rankings.

Our Solution & Winning USP

TalentMatch AI introduces a dual-engine strategic paradigm that replaces string parsing with true semantic contextual comprehension, combined with behavioral telemetry.

🧠 1. Deep Semantic Alignment

Utilizes state-of-the-art dense text vector representations (`all-MiniLM-L6-v2`) to map the holistic intent of the Job Description against the unstructured history, skills, and certifications of candidates.

Result: Catches equivalent domain skill contexts automatically, rendering keyword-stuffing transparent and ineffective.

⚡ 2. Behavioral Signal Fusing

A perfect resume on paper is useless if the candidate is passive or unavailable. Our engine extracts live, behavioral telemetry records (`redrob_signals`) directly into the mathematical scoring layer.

Result: Calculates active platform presence, engagement, and job-market readiness into the final index score.

The Engineering Architecture & Pipeline

To scale smoothly across 100,000 raw candidate records without causing memory overflow or performance lags, our system utilizes a high-efficiency **Two-Stage Retrieval Architecture**.

1. Raw Data Input
100k JSONL Records parsed & flattened

→

2. Stage 1 Filter
High-speed sub-matrix cuts pool to 2k

→

3. Vector Space
MiniLM dense text embedding matrix

→

4. Hybrid Booster
70% Semantic + 30% Active Signal

→

5. Top 100 Output
Exact formatted CSV with logic logs

Execution Metrics & Robustness: The entire 100,000 record database executes, aligns, embeds, and ranks down to the exact requested 100-row template formatting in **under 60 seconds** on a single standard cloud notebook instance.